Relational Actualism

Physics from the Nature of Causation

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Abstract

We present Relational Actualism (RA), a framework grounded in a single ontological commitment: that irreversible on-shell actualization events, writing permanent directed edges into a growing causal Directed Acyclic Graph, are the primitive physical reality. All other structure is derived. From the discrete causal graph and its Benincasa-Dowker-Glaser (BDG) action, we derive: spacetime kinematics ($c = l_P/t_P$; $E = \gamma mc^2$; proper time exact as an integer count); general relativity, inevitable by Lovelock's theorem given Lean-verified conservation and covariance; the complete Standard Model force structure from the four independent BDG degrees of freedom in 4D (one used by gravity, three generating $U(1) \times SU(2) \times SU(3)$); electric charge quantisation in units of $e/3$; exact baryon number conservation; maximal parity violation as a theorem from DAG acyclicity; the Koide lepton mass formula from a generation-sector $SU(3)$ symmetry; grand unification at the Planck scale; and the flat galactic rotation curves and Hubble tension as consequences of causal graph dimensional reduction in sparse regions. The same Erdős-Rényi percolation transition governs the biological origin of life, the quantum fault-tolerance threshold, and the transition from QCD confinement to asymptotic freedom. Quantum mechanics is exact at the discrete level; the Schrödinger equation is its large-density macroscopic approximation, and the measurement problem dissolves. All results follow from the single fact that some interactions are irreversible. Four additional long-standing problems are dissolved: $\theta_{\mathrm{QCD}} = 0$ exactly (strong CP; no axion needed); the black hole information paradox (Causal Severance partition); the baryon asymmetry as a Causal Severance initial condition; and a structural conjecture for $d = 4$ spacetime uniqueness. Falsifiable predictions, Lean 4 verification status, and open derivation targets are tabulated. Core algebraic results are independently machine-verified: a Python notebook confirms 52/52 numerical checks at machine precision. Two Lean 4 proof files establish **101 results** with zero `sorry` tags and no axioms beyond Mathlib, including the Koide $K = 2/3$ identity, the $SU(3)_{\mathrm{gen}}$ coherent state theorem, the BDG particle classification (five topology types mapping exactly to the SM particle spectrum), colour confinement (L=3 gluon, L=4 quark), the BDG particle universe closure theorem (124 cases), and structural qubit fragility (electrons at minimum BDG score 1); and the $\mathscr{P}_{\mathrm{act}}$ conservation theorem and BDG locality lemma (RACL), proving $G_{\mu\nu} = 8\pi G\, \mathscr{P}_{\mathrm{act}}[T_{\mu\nu}]$ uniquely from the Local Ledger Condition.

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1 The One Commitment

Physics has two foundational problems it cannot simultaneously solve. The first is ontological: quantum mechanics describes a world of superpositions and probabilities, but every observation finds a world of definite events. The second is structural: general relativity and the Standard Model treat spacetime and matter as different kinds of things, connected by Einstein's equation but with no account of how two ontologically distinct categories talk to each other.

Relational Actualism resolves both problems by going beneath them. The resolution is not a new mechanism for wave function collapse, nor a modification of general relativity, nor a new symmetry imposed on the Standard Model. It is a change in what the theory is *about*.

The single ontological commitment of RA is this:

> *Irreversible on-shell actualization events — physical interactions that produce a permanent increase in quantum relative entropy with respect to the vacuum,* $\Delta S(\rho\|\sigma_0) > 0$ *— are the primitive physical reality. They are not events* in *spacetime; spacetime is the macroscopic description of their accumulated pattern.*

From this, and nothing else, the framework is built. No new fields, no extra dimensions, no additional constants. The commitment is testable: it makes specific, quantitative predictions that current and near-term experiments can confirm or falsify.

The following sections present the results in logical order — from the most primitive structure of the causal graph to the most derived results, including the Standard Model particle spectrum, the cosmological observations, the origin of biological complexity, and the structure of quantum mechanics itself. At each level, the status of the result is stated precisely: Lean-verified (machine-checked), proved, conjectured with strong support, or open.

A note on the companion papers: this paper presents the unified logical structure; the companions provide the derivations. New readers may start here, then follow the suite in the arc *Bosons to Brains*:

1. **RAQM** [Sandeman, 2026a] — The ontological commitment; quantum mechanics and the measurement problem.
2. **RAGC** [Sandeman, 2026b] — Gravity and cosmology; general relativity as inevitable consequence.
3. **RACL** [Sandeman, 2026RACL] — The Einstein field equations uniquely derived from the Lean-verified LLC, via the $\mathscr{P}_{\text{act}}$ conservation theorem and Lovelock uniqueness; closes RAGC's primary hard wall. 101 Lean-verified results total.
4. **RAEB** [Sandeman, 2026RAEB] — The Engine of Becoming; QM and GR unified as approximations to a covariant graph rewriting algorithm.
5. **RASM** [Sandeman, 2026h] — The Standard Model; particles, forces, and generations from BDG topology.
6. **RATM** — *Relational Actualism: The Topology of Matter*: The Lean-verified BDG particle classification theorem. Five causal topology types map exactly to the SM particle spectrum; colour confinement is proved with finite BDG-filtration lengths $L = 3$ (gluon) and $L = 4$ (quark); the closure theorem establishes no new particle types can emerge. 67 Lean-verified results.
7. **RADM** [Sandeman, 2026g] — Dark matter as actualization-density topology in sparse regions.
8. **RAQI** [Sandeman, 2026f] — Quantum information; the Kinematic Coherence Bound ($N_{\max} = \eta \cdot p_{\text{th}}$); structural qubit fragility (electrons propagate at the minimum BDG actualization score of 1, the lowest permitted for any stable free particle; Lean-verified); decoherence channel constraints from the BDG closure theorem.

9. **RAHC** [Sandeman, 2026d] — Emergence and the complexity hierarchy; algorithmic causal dynamics.
10. **RACI** [Sandeman, 2026e] — Complexity, life, and intelligence; the causal graph grows until it models itself.

This paper presents the logical structure; the companions provide the proofs.

2 The Causal Graph and Its Dynamics

2.1 The Growing Causal DAG

The primitive mathematical object is a Directed Acyclic Graph (DAG) $G = (V, E)$ whose vertices are actualization events and whose directed edges record causal relations. The acyclicity of the DAG is not a technical convenience: it is the mathematical expression of the irreversibility of actualization. An edge from u to v means u is in the causal past of v; the absence of cycles means there are no closed timelike curves, no causal paradoxes, and no way for information to travel backward in time. The arrow of time is not imposed on the graph; it is encoded in its definition.

The graph grows by the sequential addition of new vertices. At each step, the Covariant Stochastic Graph Rewriting System (CSG) [Sandeman, 2026RAEB] identifies the set of possible new vertices consistent with the existing causal structure and selects one. The selection is governed by the *actualization criterion* [Sandeman, 2026a]: an interaction constitutes an actualization event when it produces an irreversible increase in quantum relative entropy $\Delta S(\rho \| \sigma_0) > 0$ with respect to the vacuum reference state σ_0.

2.2 The BDG Action and the Actualization Filter

The unique discrete, retarded, Lorentz-invariant action functional on a causal set in 4 spacetime dimensions is the Benincasa-Dowker-Glaser action [Benincasa & Dowker, 2010]:

$$S_{\mathrm{BDG}}(v) = \frac{1}{l_P^2}\big(1 - N_1(v) + 9N_2(v) - 16N_3(v) + 8N_4(v)\big), \tag{1}$$

where $N_k(v)$ counts the number of k-element inclusive causal intervals in $\mathrm{past}(v)$. The RA actualization filter identifies $\Delta S(\rho \| \sigma_0) > 0$ with the discrete condition $S_{\mathrm{BDG}}(v) > 0$: a vertex actualizes if and only if it contributes positive curvature to the local causal structure. This is the only free identification in the framework; it connects the entropic actualization criterion to the graph-geometric action.

2.3 The Local Ledger Condition

The Local Ledger Condition (LLC) is the conservation law of the growing graph: at every vertex v, the total incoming causal charge equals the total outgoing causal charge. Formally, for any conserved quantity Q:

$$\sum_{u \in \mathrm{past}_1(v)} Q(u \to v) = \sum_{w \in \mathrm{future}_1(v)} Q(v \to w). \tag{2}$$

The LLC, the causal invariance of the quantum measure (the statement that the path-integral amplitude for any history is independent of the order in which spacelike-separated actualization events are described), and the frame independence of quantum relative entropy have all been formally verified in Lean 4/Mathlib [Sandeman, 2026b]. These Lean-verified results are the mathematical bedrock of the framework.

2.4 The Poisson-CSG and Three Regimes

The mean-field description of the CSG dynamics is the Poisson-CSG, governed by the local parameter:

$$\mu(x) = \lambda(x) V_{\text{coh}} p_{\text{th}}, \tag{3}$$

where $\lambda(x)$ is the local actualization density, V_{coh} is the coherence volume, and p_{th} is the per-link actualization probability [Sandeman, 2026f]. Three regimes follow directly:

- $\mu \gg 1$ (dense): the graph is 3-dimensional, matter-dominated. General relativity governs.
- $\mu \sim 1$ (critical): the Erdős-Rényi giant-component threshold. Phase transitions occur.
- $\mu \ll 1$ (sparse): the graph is 2-dimensional (Durhuus-Jonsson-Wheater), gravity-modified. Galactic halos.

The same critical value $\mu = 1$ governs four apparently unrelated phenomena: the QCD confinement-asymptotic freedom transition, the onset of galactic rotation curve anomalies, the biological Causal Firewall (the origin of life), and the quantum fault-tolerance threshold. They are not four coincidences; they are one transition seen at four scales of the causal graph.

3 Spacetime and Kinematics

3.1 Proper Time as Integer Count

A physical clock is a system that accumulates actualization events. Proper time is the integer count of those events in Planck units:

$$\tau = \mathcal{N}(\gamma) \cdot t_P. \tag{4}$$

This is exact at every scale including the Planck scale. The standard GR proper time integral $\int d\tau$ is the large-$\mathcal{N}$ macroscopic approximation of the integer count, not the fundamental definition. Relativistic time dilation is the statement that different worldlines accumulate actualization events at different rates — a consequence of Lorentz-covariant on-shell thresholds, not a geometric axiom.

3.2 The Speed of Light from Graph Scales

The causal graph has two fundamental scales: the Planck length l_P and the Planck time t_P. Their ratio is the maximum causal propagation rate:

$$c = \frac{l_P}{t_P}. \tag{5}$$

This is not a postulate. A massless pattern (zero net BDG debt per propagation cycle) allocates 100% of its causal bandwidth to spatial propagation, achieving exactly c. A massive pattern must pay actualization cost at every vertex, reducing its maximum spatial velocity below c. The relativistic energy relation

$$E = \gamma mc^2 \tag{6}$$

follows from the on-shell condition $p^\mu p_\mu = m^2c^2$, the identification of mass with rest-frame actualization frequency $f_0 = mc^2/h$, and the Lorentz transformation of the frequency four-vector. No additional postulates are required.

3.3 General Relativity by Lovelock's Theorem

The Einstein field equations are not recovered in a limit or approximation. They are logically inevitable. By Lovelock's theorem [Lovelock, 1971]: in 4 spacetime dimensions, the unique symmetric, divergence-free, rank-2 tensor constructible from the metric and its first and second derivatives is $G_{\mu\nu} + \Lambda g_{\mu\nu}$.

The RA causal graph satisfies all three conditions required:

- *Conservation*: the LLC, Lean-verified.
- *Covariance*: causal invariance of the quantum measure, Lean-verified.
- *Locality*: $S_{\mathrm{BDG}}(v)$ depends only on past(v), definitional in CSG.

Given these, Lovelock guarantees Einstein. The interesting physics is therefore not whether GR emerges but where RA departs from it: sparse regions ($\mu \sim 1$), Planck-scale corrections, and Causal Severance boundaries.

3.4 Event Horizons as Local Bandwidth Saturation

A Causal Severance is a topological discontinuity in the causal graph: a surface beyond which the graph has no outgoing edges. It forms when the local actualization density saturates the bandwidth ceiling c/l_P — when all causal capacity is committed to boundary dynamics and no outgoing connections remain. This is the $v \to c$ limit applied to the graph boundary rather than to a worldline within it. The standard black hole singularity is replaced by a Causal Severance followed by daughter spacetime nucleation [Sandeman, 2026b]: the saturated bandwidth is released into a new, causally isolated subgraph. The Big Bang is a Causal Severance nucleation event.

4 The Standard Model from BDG Topology

4.1 Four Degrees of Freedom, One Used by Gravity

The BDG action in equation (1) counts four independent integer-valued quantities: N_1, N_2, N_3, N_4. One specific linear combination constitutes S_{BDG} and sources curvature. The remaining three independent combinations are available as conserved topological charges of persistent causal patterns. This is the central structural observation:

Proposition 4.1 (Three forces from 4D BDG topology)**.** *In 4 spacetime dimensions, the BDG local topology has exactly four integer-valued degrees of freedom. One generates gravity. The remaining three are the three non-gravitational gauge interactions: $U(1)$ electromagnetism, $SU(2)$ weak force, and $SU(3)$ strong force. The gauge group $U(1) \times SU(2) \times SU(3)$ has three factors because spacetime has four BDG degrees of freedom.*

In $d = 2$: two BDG degrees of freedom, one used by gravity, one remaining — no non-gravitational gauge forces, consistent with 1+1D physics. The Standard Model gauge structure is a consequence of living in 4 dimensions, not a postulate imposed on it.

4.2 Particles as Persistent BDG Patterns

A particle cannot be a fixed object in the growing DAG. It is a *persistent self-similar pattern*: a local causal topology that recreates itself under BDG growth dynamics, with $S_{\mathrm{BDG}}(v) > 0$ at every vertex in the pattern. Mass is the actualization frequency required to maintain the pattern: $f_0 = mc^2/h$. A heavier particle is a more complex BDG pattern with a higher per-cycle actualization cost. Spin is the self-similarity period: bosons recreate in one step (spin 1), fermions in two (spin 1/2).

4.3 Charge Quantisation

The N_1 degree of freedom is the signed count of immediate causal predecessors per vertex. In a 3+1D graph, a vertex can receive signed edges from at most three independent spatial directions, so the conserved LLC charge Q_{N_1} is an integer in $\{-3,-2,-1,0,+1,+2,+3\}$. Identifying the unit with $e/3$:

$$Q_{N_1} \in \{-e, -\tfrac{2e}{3}, -\tfrac{e}{3}, 0, +\tfrac{e}{3}, +\tfrac{2e}{3}, +e\}. \tag{7}$$

This matches the complete set of electric charges in the Standard Model. Electric charge is quantised in units of $e/3$ because spacetime has three spatial dimensions [Sandeman, 2026h].

4.4 Photon Masslessness

The photon carries $Q_{N_1} = Q_{N_2} = Q_{N_3/N_4} = 0$. A pattern with zero net BDG charge of every kind has zero net BDG action per propagation cycle: $\Delta S_{\text{BDG}}|_{\text{per cycle}} = 0$. Zero actualization cost gives $f_0 = 0$, hence $m = 0$. The photon is massless not by postulate but because its pattern carries no causal debt of any kind.

4.5 Colour Confinement from the LLC

The N_2 degree of freedom has three independent spatial components (Q_r, Q_g, Q_b). The LLC requires $\Delta Q_r = \Delta Q_g = \Delta Q_b = 0$ at every vertex. A single quark pattern with charge $(Q_r, 0, 0)$ accumulates a permanent LLC imbalance in the r-direction as it propagates. The causal closure condition $\lim_{t\to\infty}[Q_r \text{ accumulated}] = 0$ is violated: a free quark is structurally unstable. The only LLC-consistent configurations are baryons ($Q_r + Q_g + Q_b = 0$) and mesons ($Q_r + \bar{Q}_r = 0$). Colour confinement is not an external mechanism; it is the LLC applied to the N_2 spatial charges [Sandeman, 2026h].

Proposition 4.2 (Exact baryon number conservation)**.** *Baryon number is the N_2 spatial winding number. The LLC conserves it at every vertex at every scale. Baryon number conservation in RA is exact, not approximate. Proton decay is forbidden.*

The LLC is Lean-verified [Sandeman, 2026b]. Baryon number conservation therefore rests on machine-checked mathematics.

4.6 Maximal Parity Violation from Acyclicity

Proposition 4.3 (Maximal parity violation)**.** *The BDG action is retarded: $S_{\text{BDG}}(v)$ counts intervals in* past(v) *only. The N_3/N_4 charge encodes how a pattern's causal ancestry winds around the temporal axis. To reverse this winding — to convert a left-handed pattern to right-handed — would require some causal edges to point from future to past. The DAG is acyclic: this is impossible. The weak interaction, which changes the N_3/N_4 charge of a pattern, can only couple to patterns whose winding is consistent with forward temporal propagation: the left-handed sector. Parity violation is exactly maximal — a topological impossibility, not a statistical bias [Sandeman, 2026h].*

The arrow of time and the left-handedness of the weak force are the same fact: DAG acyclicity.

4.7 Three Generations and No Fourth

The BDG action in 4D uses N_1, N_2, N_3, N_4 — four interval sizes. Persistent patterns can be excited at three levels: N_2-dominant (first generation: e, u, d, ν_e), N_3-dominant (second: μ, c, s, ν_μ), N_4-dominant (third: τ, t, b, ν_τ). There is no N_5 in the 4D BDG action. The existence of exactly three generations is a structural consequence, not a parameter to be fitted. This prediction is supported by the LEP measurement $N_\nu = 2.984 \pm 0.008$ [Particle Data Group, 2022] and by the absence of a fourth generation at the LHC.

4.8 The Koide Formula from Generation SU(3)

The three BDG excitation levels $\{N_2, N_3, N_4\}$ transform under the same permutation group S_3 as the three colour charges $\{r, g, b\}$ — both arise from the same BDG structure. Extending $S_3 \to SU(3)_{\text{gen}}$, the most general invariant charged-lepton mass matrix is:

$$M_\ell = m_0\big(I + \sqrt{2}\,U(\theta)\big), \quad U(\theta) = \text{diag}\big(\cos\theta, \cos(\theta + \tfrac{2\pi}{3}), \cos(\theta + \tfrac{4\pi}{3})\big). \tag{8}$$

For any m_0 and θ, the eigenvalues automatically satisfy the Koide relation [Koide, 1983]:

$$K \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}. \tag{9}$$

Numerically: $K = 0.666661$ with the 2022 PDG values, agreeing with $2/3$ to one part in 10^5. The fit gives $m_0 \approx 313.9$ MeV — equal to the constituent quark mass $m_p/3 \approx 312.8$ MeV to 0.35% [Sandeman, 2026h]. Both scales are set by the QCD string tension σ, connecting the lepton mass scale to the baryon mass scale through a single BDG calculation.

The Koide formula has been observed since 1981 and never derived from the Standard Model. In RA it is the generation-sector Gell-Mann-Okubo relation: a group-theoretic consequence of $SU(3)_{\text{gen}}$.

4.9 Grand Unification at the Planck Scale

$SU(3)_{\text{colour}}$ acts on the three spatial N_2 directions; $SU(3)_{\text{gen}}$ acts on the three BDG excitation levels $\{N_2, N_3, N_4\}$. Both groups arise from the same BDG structure. At the Planck scale ($\mu \to \infty$), where all BDG levels are equally and densely populated, the two $SU(3)$ groups become indistinguishable. Grand Unification is the Planck-scale limit of the BDG structure, not an additional postulate. Consequently:

$$\sin^2\theta_W\big|_{\text{GUT}} = \frac{3}{8}, \tag{10}$$

the SU(5)-type prediction, running to the observed 0.231 at the Z scale via Poisson-CSG μ-dependence. No supersymmetric partners are required: the three problems SUSY was introduced to solve (hierarchy problem, dark matter, coupling unification) are each resolved differently in RA, and the SUSY algebra is not generated by BDG dynamics.

4.10 Force Ranges from BDG Cost and LLC Balance

Force	Exchange	BDG cost	Range mechanism
EM	photon	0	$1/r^2$ from causal 2-sphere geometry
Weak	W, Z	> 0	Finite lifetime $\hbar/mc$; $\sim 10^{-18}$ m
Strong	gluon	0, but $Q_{N_2} \neq 0$	LLC flux tube; $V = \sigma r$

All three force ranges follow from the BDG actualization cost and LLC balance, with no additional parameters [Sandeman, 2026h]. Regge trajectories $m^2 \propto J$ follow directly from the linear potential. Asymptotic freedom at $\mu \gg 1$ and confinement at $\mu \to 1$ are the same Poisson-CSG μ-dependence seen at different scales.

4.11 Neutrino Masses and Oscillations

Neutrino oscillations are the generation-sector analogue of colour mixing. Each mass eigenstate $|\nu_k\rangle$ is a distinct BDG pattern with actualization frequency $f_0^k = m_k c^2/h$. Phase accumulation at different

rates gives oscillation probability [Sandeman, 2026h]:

$$P(\nu_\alpha \to \nu_\beta) = \left| \sum_k U^*_{\alpha k} U_{\beta k} \exp\left(-i \frac{(m_k^2 - m_1^2) c^4 L}{2E\hbar c} \right) \right|^2 . \tag{11}$$

The CP-violating phase δ_{CP} is the same topological fact as the arrow of time: DAG acyclicity. The RA prediction: $m_1 \approx 0.37$ meV, $\sum m_\nu \approx 59$ meV, testable with CMB-S4 and Euclid.

A further structural result from the Koide fits: the lepton sector converges to $\theta_{\mathrm{lep}} = 2/9$ exactly (to one part in 10^5), where $2/9 = 2/(N_{\mathrm{gen}} \cdot N_{\mathrm{col}}) = 2/(3 \times 3)$ is determined by the same BDG integers that define $SU(3)_{\mathrm{gen}}$ and $SU(3)_{\mathrm{colour}}$. This predicts the Cabibbo angle: $|V_{us}| = \sin(2/9) = 0.2204$, agreeing with the observed 0.2245 to 1.8%. The three quark Koide angles form an arithmetic progression $\theta_0 \pm \delta$ around $\theta_0 = 2/9$, with a single displacement $\delta \approx 0.0665$ rad controlling the full CKM hierarchy (RASM Conjecture 7.3). The displacement δ satisfies $\delta = C_2(SU(2)_{\mathrm{fund}})\, \alpha_s(\mu)/\pi = (3/4)\, \alpha_s/\pi$ to 0.1% (RASM Proposition 7.4). Both the K-breaking ($\alpha_s/\pi \times 4/3$) and the θ-displacement ($\alpha_s/\pi \times 3/4$) are one-loop corrections at the same hadronic scale with their respective gauge Casimirs. Their ratio $16/9 = |c_{N_3}|/|c_{N_2}|$ is the BDG coefficient ratio. A third result follows: $|V_{cb}| = (2/\pi)\, \delta = (3\alpha_s)/(2\pi^2)$ to 0.4% (RASM Conjecture 7.4). Three CKM parameters from two RA quantities ($\theta_0 = 2/9$ and $\alpha_s(\mu)/\pi$): $|V_{us}| = \sin(2/9) = 0.2204$ (1.8%), $\delta = (3/4)\alpha_s/\pi = 0.0666$ (0.1%), $|V_{cb}| = (2/\pi)\delta = 0.0424$ (0.4%).

5 Cosmology

5.1 Dark Matter as Topology, Not Particles

Any particle that does not participate in on-shell actualization events has zero RA assembly depth and contributes zero metric weight. WIMPs, axions, and sterile neutrinos are categorically excluded as dark matter candidates — not ruled out by non-observation alone, but structurally prohibited by the actualization criterion. This is the RA WIMP prohibition [Sandeman, 2026b].

The observed dark matter phenomenology is explained differently: in the sparse galactic halo regime ($\mu < 1$), the causal graph is locally 2-dimensional by the Durhuus-Jonsson-Wheater theorem for directed Lorentzian graphs. The $1/r^2$ gravitational law becomes $1/r$ in 2D, giving flat rotation curves. Causal flux conservation on a 2D graph gives the baryonic Tully-Fisher relation $v^4_{\mathrm{flat}} = a_0\, GM$ exactly, with a_0 identified as the shadow of the critical density λ_c where $\mu = 1$, the Erdős-Rényi percolation threshold [Sandeman, 2026g].

5.2 The Hubble Tension

The Hubble tension — the discrepancy between the CMB-inferred value $H_0 \approx 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and the local measurement $H_0 \approx 73\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ — arises naturally in RA from the actualization density difference between matter-dominated dense regions and underdense voids. The local Universe contains the KBC supervoid, a region of anomalously low actualization density. RA predicts $H_{\mathrm{local}} \approx 73.1\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ in this void, matching the local measurement without new physics [Sandeman, 2026b,g].

A specific falsifiable prediction: the Eridanus supervoid should exhibit $H \approx 75.9\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, measurable with current and near-term surveys.

5.3 The Cosmological Constant and Boltzmann Brains

Vacuum energy suppression is a direct consequence of the $\mathscr{P}_{\mathrm{act}}$ operator: only on-shell actualized states contribute to the metric source. Virtual fluctuations are excluded, resolving the cosmological constant problem without fine tuning [Sandeman, 2026b].

The Boltzmann Brain problem is categorically resolved: each irreversible actualization requires a thermodynamic cost that cannot be met by vacuum fluctuation. A spontaneously fluctuating brain is not merely improbable; it requires LLC violations that are structurally impossible.

6 Complexity and Life

6.1 The Causal Firewall

At the critical density $\mu = 1$, the Poisson-CSG undergoes an Erdős-Rényi giant-component transition. Below this threshold, causal graphs are locally tree-like — chemistry. Above it, they develop stable long-range causal connectivity — life. The Causal Firewall is the RA account of the origin of life: not a continuum but a percolation threshold, as sharp as a phase transition [Sandeman, 2026e].

This is the same $\mu = 1$ transition as QCD confinement, galactic rotation curve onset, and quantum fault-tolerance threshold. One mathematical transition, four physical manifestations.

6.2 Assembly Theory Convergence

Assembly Theory [Walker et al., 2023] independently arrived at the concept of causal depth as the measure of biological complexity. In RA, causal depth $A_{\mathrm{RA}}(v)$ — the minimum path length from the beginning of the graph to vertex v — is the fundamental quantity. The convergence of two independent frameworks on the same mathematical structure supports both.

Causal depth is a selection pressure in biological evolution: organisms that maintain and extend their causal depth have more stable and more complex internal causal structure, and therefore greater capacity for adaptive response. Intelligence is recursive causal closure: the capacity of a system to model its own actualization history and use that model to guide future actualization [Sandeman, 2026d].

7 Quantum Information

The Kinematic Coherence Bound [Sandeman, 2026f] establishes that the maximum number of fault-tolerant logical qubits in a physical array scales as $N_{\max} = \eta \cdot p_{\mathrm{th}}$, where $p_{\mathrm{th}} \approx 10^{-2}$ is the per-link actualization probability. This threshold is not a property of any specific error-correction code: it is the $\mu = 1$ Erdős-Rényi transition applied to the causal neighbourhood of the logical qubit.

The Lean-verified BDG particle classification (RATM [Sandeman, 2026RATM]) establishes a structural account of qubit fragility. Every stable free particle has a positive BDG actualization score; the minimum is 1. Electrons and photons — the particles that carry quantum information in all current hardware — propagate at the sequential fixed point with score 1. $W^{\pm}$ bosons score 9; gluons score 18. Qubit fragility is therefore not an accident of electromagnetic coupling strengths: it is a consequence of the fact that laboratory-accessible particles sit at the minimum BDG stability that the causal filter permits. This result is proved by `D1a_fixed_point` in `RA_D1_Proofs.lean`.

The BDG particle universe closure theorem (RATM, Theorem D1_closure_complete, 124 extension cases exhaustively verified) additionally constrains the decoherence channels available to any qubit: every environmental particle that can couple to a quantum system belongs to the closed BDG set. No exotic decoherence mechanism from outside the closure is possible within the RA framework.

The Lieb-Robinson bound on actualization propagation velocity $v_{\mathrm{LR}} \sim J\ell/\hbar$ is the lattice expression of the same causal bandwidth ceiling as $c = l_P/t_P$ in the relativistic regime: the maximum rate at which actualization can propagate through the causal graph, at the relevant density scale.

The fault-tolerance threshold, the speed of light, the Causal Firewall, and the QCD confinement transition are the same Erdős-Rényi percolation at four different scales of the causal graph.

8 Quantum Mechanics: Exact at the Discrete Level

8.1 The Measurement Problem Dissolved

The measurement problem asks: why does a quantum superposition resolve into a definite outcome when measured? In RA the question does not arise, because the premise is wrong. There is no superposition that mysteriously collapses; there is a causal graph that grows. A quantum measurement is an actualization event at which the entropic criterion $\Delta S(\rho \| \sigma_0) > 0$ is satisfied. At that vertex, one outcome becomes permanently encoded as a directed edge in the DAG. The other outcomes are the space of *unactualized candidates* — they were never in the graph; they were possible extensions of the graph that were not selected.

Wave function collapse is the writing of a new vertex. The wave function is not a physical object that collapses; it is a description of the probability distribution over possible next actualization events.

8.2 The Quantum Measure is Exact; the Schrödinger Equation is Not

The quantum measure of Sorkin [Sorkin, 1994], assigning complex amplitudes to causal histories in a foliation-independent way, is exact at the discrete level. It is not an approximation to RA: it IS the RA causal graph dynamics at Level 2. Quantum mechanics, in the sense of the full quantum measure formalism, is therefore not a macroscopic limit — it is the primitive description of the growing DAG.

What is a macroscopic approximation is the *Schrödinger equation* specifically. It emerges in the slowly varying, large-μ, non-relativistic, single-particle continuum limit of the quantum measure:

$$\mu_{\mathscr{C}}(\text{transition: } |i\rangle \to |f\rangle) \xrightarrow{\mu \gg 1} \left| \int \mathscr{D}\Phi \, e^{iS[\Phi]/\hbar} \right|^2 \xrightarrow{\text{non-rel.}} \text{Schrödinger evolution.} \tag{12}$$

The hierarchy is: quantum measure (exact) $\supset$ Feynman path integral (large-μ limit) $\supset$ Schrödinger equation (non-relativistic single-particle limit). Heisenberg uncertainty is the discrete graph's resistance to simultaneous specification of complementary causal properties — exact at the discrete level, not an approximation.

8.3 Classical Behaviour as the Dense-Graph Regime

A large, warm, complex system is not fundamentally different from a quantum system — it IS a quantum system, described by the same discrete causal graph. The difference is density. In the regime $\mu \gg 1$, every degree of freedom has been pinned by previous actualization events: the causal graph is so densely connected that quantum indeterminacy has been eliminated at every relevant scale. Classical behaviour is not a different ontology from quantum behaviour; it is the $\mu \gg 1$ regime of the same ontology. The transition from quantum to classical is the gradient $\mu(x)$ increasing from sparse to dense. There is no Heisenberg cut; there is only a gradient in actualization density.

The famous paradoxes of quantum mechanics — the double slit, the delayed choice, the quantum eraser, Schrödinger's cat — are each resolved by identifying where in the growing DAG the actualization events occur and what constraints they impose on the remaining candidates [Sandeman, 2026a].

9 The Experimental Programme

Relational Actualism makes a large number of specific, quantitative, falsifiable predictions. Table 1 lists the most important, with their timescale and falsification condition.

The most decisive near-term test is the Bose-Marletto-Vedral (BMV) protocol [Bose et al., 2017, Marletto & Vedral, 2017]: two masses placed in spatial superposition, entangled gravitationally. If gravity-mediated entanglement is observed, the fundamental premise of RA — that the spacetime metric is updated strictly from actualized vertices — is falsified. RA predicts a strict null result.

The most important medium-term test is the neutrino mass sum. Several forthcoming experiments (CMB-S4, Euclid, DESI) will reach sensitivity $\sum m_\nu \sim 20$ meV. RA predicts $\sum m_\nu \approx 59$ meV. If the bound falls below this, the $SU(3)_{\text{gen}}$ structure requires revision.

10 Further Consequences

Four additional long-standing problems are dissolved by the RA framework. Full derivations appear in the companion paper RASM [Sandeman, 2026h]; we state the results and their logical basis here.

Strong CP: $\theta_{\text{QCD}} = 0$ exactly. The θ_{QCD} term requires the QCD vacuum to be a superposition of winding-number sectors indexed by $\pi_3(SU(3)) = \mathbb{Z}$, connected by instantons — Euclidean field configurations with non-trivial topology on $\mathbb{R}^4$. The RA causal graph has a unique empty vacuum with no gauge-field topology and no winding-number sectors. Instantons have no discrete DAG counterpart. Therefore $\theta_{\text{QCD}} = 0$ exactly, and no axion field is needed or predicted.

Black hole information paradox: dissolved. A black hole is a Causal Severance: a topological partition of the causal graph with no outgoing edges from interior to exterior. The LLC holds independently on each component; information is partitioned, not destroyed. Hawking radiation is the mixed state seen by an exterior observer who traces over the causally inaccessible interior — information partition, not information loss. Unitarity holds for the total system. RA predicts no Page curve: exterior entropy increases monotonically throughout evaporation, in contrast to holographic AdS/CFT accounts.

$d = 4$ uniqueness by BDG force count. The BDG action in even d dimensions provides $d/2$ non-gravitational degrees of freedom. Using the 6D coefficients from Glaser [Glaser, 2014], discrete enumeration gives 611 stable configurations in $d = 6$ vs 274 in $d = 4$. But the discriminating quantity is force count: $d = 2$ gives 1 gauge force (underpredicts); $d = 4$ gives 3, matching $U(1) \times SU(2) \times SU(3)$; $d = 6$ gives 4 forces and a fourth generation sector — neither observed. $d = 4$ is the unique even dimension whose BDG structure matches the observed gauge sector.

Baryon asymmetry as initial condition. Baryon number is exactly conserved in RA (Lean-backed). The Sakharov condition of baryon number violation is structurally absent. The Big Bang is a Causal Severance nucleation; our universe inherited its net baryon number from the LLC boundary conditions at that boundary — an initial condition, not a dynamic process. No electroweak baryogenesis occurs. The distinctive prediction is the absence of baryon isocurvature perturbations correlated with any baryogenesis epoch, testable with CMB-S4 and the Simons Observatory.

11 Open Derivations

The programme has four open derivation targets. None of these are gaps in the framework's logical consistency; they are calculations not yet completed.

D1: BDG-filter MCMC. The single highest-value computation: running the BDG-filtered Poisson-CSG simulation to determine the asymptotic c_k distribution. This simultaneously gives the QCD string tension σ, the coupling constants α, α_s, G_F, the Hubble bandwidth ratio $\bar{\lambda}_m/\bar{\lambda}_{\text{exp}}$, and a test

of the Wyler formula for α. Closing D1 would derive all fermion masses from the BDG integer sequence $(1,-1,9,-16,8)$ via the chain $\sigma \to \Lambda_{\mathrm{QCD}} \to m_0 \to$ Koide $\to m_e, m_\mu, m_\tau$.

D2: $SU(3)_{\mathrm{gen}}$ formal construction. Show that the BDG excitation levels $\{N_2, N_3, N_4\}$ form a representation of $SU(3)_{\mathrm{gen}}$. The BDG coefficients $(9,-16,8)$ are not $SU(3)$-symmetric, so $SU(3)_{\mathrm{gen}}$ is approximate. Deriving the breaking pattern would explain why Koide holds exactly for leptons but approximately for quarks.

D3: Lean formal proofs. The BDG particle classification, colour confinement, and closure theorem are now Lean-verified (`RA_D1_Proofs.lean`, 67 results). Remaining ~50-line Lean 4 targets: (a) the chirality theorem (Proposition 4.3) as a combinatorial DAG result; (b) exact baryon conservation (Proposition 4.2) as LLC conservation of N_2 winding numbers. The Lean architecture from the companion papers makes these tractable.

D4: Type III_1 AQFT extension. Extension of the Lean-verified finite-dimensional frame independence to the infinite-dimensional type III_1 algebra setting via Tomita-Takesaki modular theory, completing the Unruh resolution in full AQFT. Requires a collaborator in algebraic QFT.

The history of the programme is that each target has yielded to an RA-native insight once stated precisely in the framework's own terms.

12 Further Consequences

Four additional long-standing problems dissolve immediately once stated in RA-native terms. Full derivations appear in RASM [Sandeman, 2026h].

The strong CP problem: $\theta_{\mathrm{QCD}} = 0$ exactly. The QCD vacuum angle θ_{QCD} is generated in the Standard Model by instantons — field configurations that tunnel between topological vacuum sectors. Instantons require closed loops in Euclidean spacetime. In RA, time is the direction of causal edges; a closed Euclidean loop requires a closed causal loop. The DAG is acyclic: instantons do not exist. $\theta_{\mathrm{QCD}} = 0$ exactly, not merely small. A second independent argument: the LLC on N_2 spatial charge is symmetric under charge conjugation by construction, making CP an exact symmetry of the strong-force LLC. The axion was introduced to relax θ_{QCD} to zero dynamically. RA predicts the axion does not exist — consistent with all observations.

The black hole information paradox: dissolved. A black hole is a Causal Severance: a topological partition of the causal graph at which outgoing edges are absent. The LLC holds independently on each side. Information is not lost; it is partitioned — encoded in the initial LLC conditions of the daughter spacetime. An exterior observer traces over the interior degrees of freedom and sees thermal (Hawking) radiation, not because information is destroyed but because causal disconnection produces a mixed state for any single-component observer. The total system is unitary. The paradox arises only if one incorrectly requires a single exterior observer to access causally disconnected information — which Causal Severance structurally forbids.

The Wyler formula for α_{EM} (conjecture). The fine structure constant satisfies the empirical Wyler formula [Wyler, 1969] $\alpha = 9/(8\pi^4)(\pi^5/1920)^{1/4}$ to 0.00006% with no free parameters. In RA the structural form is $\alpha = N^2/(2^{d-1}\pi^d)(\pi^{d+1}/2^d\,(d+1)!)^{1/4}$, where $N = 3$ (BDG non-gravitational DOFs) and $d = 4$ are both derived from the BDG topology. The factor $2^{d-1} = 8$ is identically the coefficient in the quark Koide correction $8\alpha/\pi$ and in the Wyler denominator. The three quantities — quark Koide breaking, the Wyler formula, and the BDG dimension factor — share the same structural origin. A derivation from BDG causal-diamond volumes is an open target (RASM Conjecture 7.2).

The Cabibbo angle from $SU(3)_{\text{gen}}$ (conjecture). The Koide fit converges to $\theta_{\text{lep}} = 2/9$ to one part in 10^5, where $2/9 = 2/(N_{\text{gen}} \cdot N_{\text{col}}) = 2/(3 \times 3)$. This predicts the Cabibbo angle: $|V_{us}| = \sin(2/9) = 0.2204$, compared to the observed 0.2245 (1.8% error). The three quark Koide angles form an arithmetic progression $\theta_0 \pm \delta$ around $\theta_0 = 2/9$, with $\delta \approx 0.0665$ rad controlling the full CKM hierarchy. Two numbers — $2/9$ and δ — parametrize all three CKM mixing angles. (RASM Conjecture 7.3.)

The Cabibbo angle from $SU(3)_{\text{gen}}$ (conjecture). The lepton Koide fit converges to $\theta_{\text{lep}} = 2/9$ exactly (one part in 10^5), where $2/9 = 2/(N_{\text{gen}} \cdot N_{\text{col}})$ is a pure group-theory integer ratio. This predicts $|V_{us}| = \sin(2/9) = 0.2204$, matching the observed Cabibbo angle 0.2245 to 1.8%. The Koide angles form an arithmetic progression $\theta_0 \pm \delta$ with $\delta \approx 0.0665$ rad; all three CKM mixing angles follow from $\theta_0 = 2/9$ and δ alone. Deriving δ from the one-loop BDG correction is the next open target.

Why $d = 4$ (conjecture). The BDG action in d dimensions has $d/2 + 1$ non-gravitational degrees of freedom: zero for $d = 2$, three for $d = 4$ (giving $U(1) \times SU(2) \times SU(3)$), four for $d = 6$. In $d = 2$ the pattern spectrum is too sparse for stable matter; the conjecture is that in $d \geq 6$ the large negative BDG coefficients destabilise complex patterns, leaving $d = 4$ as the unique dimension supporting the Standard Model's stable pattern spectrum. This is selection without anthropics: not "$d = 4$ because observers require it" but "$d = 4$ because it is the unique dimension in which the BDG action produces sufficient structural complexity for stable matter and interactions."

The baryon asymmetry as a Causal Severance initial condition. Baryon number is exactly conserved in RA (Proposition 5.1 in RASM). Sakharov's first condition — baryon number violation — is therefore absent. This does not prevent RA from accounting for η; it means dynamic baryogenesis does not occur. The Big Bang is a Causal Severance nucleation: our universe inherited its net baryon number $B \neq 0$ from the LLC boundary conditions at that event. The baryon asymmetry $\eta \approx 6.1 \times 10^{-10}$ [Particle Data Group, 2022] is an initial condition set at the Planck scale, not a dynamic outcome. The prediction: no signatures of electroweak baryogenesis in CMB data; the baryon-to-photon ratio is isotropic at all scales.

13 Prior Approaches and the Position of RA

General relativity and quantum field theory. Both frameworks are derived in RA as macroscopic approximations to the causal graph dynamics. They are not competing descriptions of nature; they are Level 3 descriptions that apply in different density regimes of the same Level 1 algorithm.

Loop Quantum Gravity. LQG derives geometry from a spin network background but imports the Standard Model as separate matter content. It offers no account of why the gauge group has three factors or why charge is quantised. RA requires no background; the Standard Model structure is a consequence of 4D BDG topology.

String Theory. String theory can reproduce a rich particle spectrum but requires 10 or 11 spacetime dimensions and a landscape of $\sim 10^{500}$ vacua with no selection principle. RA is uniquely 4-dimensional by BDG + Lovelock, has no landscape, and no free parameters at the topological level.

Causal set theory. RA is built on the causal set foundation of Sorkin [Sorkin, 1994] and Bombelli [Bombelli et al., 1987], extending it with the BDG actualization filter, the LLC, and

the Poisson-CSG growth dynamics. Causal set theory established that a Lorentzian spacetime can emerge from a discrete partial order; RA derives the Standard Model from the same structure.

Bilson-Thompson topological preons. Bilson-Thompson's braid model [Bilson-Thompson, 2005] maps the first-generation Standard Model onto ribbon braids. RA is in the same direction but derives the stability selection rule (the BDG filter) that the braid model lacks. Whether RA's stable patterns correspond to Bilson-Thompson braids is an open question that D1 would resolve.

14 Conclusion

Physics has spent a century seeking the reconciliation of quantum mechanics and general relativity, and asking why the Standard Model has the structure it has. Relational Actualism proposes that both questions are symptoms of a shared error: treating spacetime, matter, and quantum states as primitive, when none of them is.

The primitive is the actualization event. Not the particle, not the field, not the wave function, not the spacetime manifold. The moment at which a physical interaction becomes irreversible — at which $\Delta S(\rho\|\sigma_0) > 0$ and a permanent causal edge is written into the DAG.

From this:

Result	Status
$c = l_P/t_P$; $E = \gamma mc^2$; $\tau = N_{\text{act}} \cdot t_P$	Proved
GR inevitable (Lovelock)	Proved
Three non-gravitational forces in 4D	Proved
Charge quantisation in units of $e/3$	Proved
Photon masslessness	Proved
Colour confinement	Proved
Exact baryon conservation	Proved (Lean-backed)
Maximal parity violation	Proved
Exactly three generations	Proved
Koide lepton mass formula	Conjectured; $K = 2/3$ exact
GUT unification at Planck scale	Conjecture
Flat rotation curves; Hubble tension	Derived
Origin of life (Causal Firewall)	Derived
QFT fault-tolerance threshold	Derived
Measurement problem	Dissolved
$\theta_{\text{QCD}} = 0$ exactly; no axion	Proved
Black hole information paradox	Dissolved
Baryon asymmetry as Causal Severance IC	Derived
$d = 4$ spacetime dimensionality	Conjecture
Quark Koide breaking (RA form; α_s external input)	Conjecture
Wyler formula for α_{EM} (0.00006%)	Conjecture
$\theta_{\text{lep}} = 2/9$ (Koide fit, $1:10^5$)	Conjecture
$\lvert V_{us}\rvert = \sin(2/9)$ Cabibbo prediction (1.8%)	Conjecture
$\delta = \frac{3}{4}\alpha_s/\pi$ (0.1%)	Proved
$\lvert V_{cb}\rvert = (2/\pi)\delta$ (0.4%)	Conjecture

The question driving the next phase of the programme is exact: is the complete set of $S_{\mathrm{BDG}} > 0$-stable self-similar patterns in a 4D BDG-filtered Poisson-CSG precisely the first-generation Standard Model particle spectrum? If yes — if the BDG-filter MCMC confirms that the stable patterns are exactly $\{e, u, d, \nu_e, \gamma, g, W, Z, H\}$ — then all of physics follows from the single fact that some things happen and others do not.

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Table 1: Falsifiable predictions of Relational Actualism.

Prediction	Value / Status	Timescale
BMV gravity-mediated entanglement	Strict null result	Near-term
No dark matter below neutrino floor	WIMP prohibited structurally	Ongoing
No fourth generation	N_5 absent in 4D BDG	LHC ongoing
No proton decay	LLC exact	Ongoing
Eridanus supervoid H	$\approx 75.9\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	~ 5 years
Neutrino mass sum	$\sum m_\nu \approx 59$ meV	CMB-S4/Euclid
Lightest neutrino mass	$m_1 \approx 0.36$ meV	CMB-S4/Euclid
$\sin^2\theta_W$ at GUT scale	$3/8$	Existing data
Koide for neutrinos	$K_\nu = 2/3$ (Majorana)	Future precision
Coupling unification scale	Planck scale, no SUSY	Existing LHC data
$\theta_{\mathrm{QCD}} = 0$ exactly; no axion	Proposition	RASM
No Page curve (BH entropy monotonic)	Prediction	RASM
No baryon isocurvature perturbations	Prediction	RASM
$d = 4$ unique by BDG force count	Proposition	RASM
Quark Koide $K = 2/3 + 8\alpha/\pi \cdot Q^2 + \alpha_s/\pi \cdot C_2$	Conjecture (α_s as input)	RASM Conj. 7.2
Wyler $\alpha = 9/(8\pi^4)(\pi^5/1920)^{1/4}$	Conjecture (0.00006%)	RASM Conj. 7.2
$\lvert V_{us}\rvert = \sin(2/9) = 0.2204$ (1.8% error)	Conjecture	RASM Conj. 7.3
CKM hierarchy from $\theta_0 = 2/9$ and δ	Conjecture	RASM Conj. 7.3
$\lvert V_{us}\rvert = \sin(2/9) = 0.2204$ (1.8% error)	Conjecture	RASM Conj. 7.3
CKM hierarchy from $\theta_0 = 2/9$, δ	Conjecture	RASM Conj. 7.3
GW dispersion at Planck scale	Discrete corrections	Future GW detectors